

Nutritional intake of ω -3 fatty acid intake and clinical grade of prostate cancer

Nagi B. Kumar^{a,b,c,*}, Saira Bahl^a, Daniel Lemay^b, Jasreman Dhillon^{a,c}, Michael Poch^d, Brandon Manley^d, Roger Li^d, Julio Pow-Sang^d, Alice Yu^d, Junmin Whiting^e, Michael J. Schell^e

^a Cancer Epidemiology Program, Moffitt Cancer Center and Research Institute, Tampa, FL 33612, United States

^b University of South Florida Morsani College of Medicine, Moffitt Cancer Center and Research Institute, Tampa, FL 33612, United States

^c Cancer Epidemiology Program, Pathology, Moffitt Cancer Center and Research Institute, Tampa, FL 33612, United States

^d Genitourinary Oncology, Moffitt Cancer Center and Research Institute, Tampa, FL 33612, United States

^e Biostatistics & Bioinformatics, Moffitt Cancer Center and Research Institute, Tampa, FL 33612, United States

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ABSTRACT

Background: Recent laboratory and some human studies have shown that ω -3 fatty acids (FA) can inhibit tumor cell growth and induce a local anti-tumor inflammatory response, independently of androgen levels in prostate cancer (PCa) models. Our objective was to conduct a cohort study to evaluate if PCa patients with higher intake of ω -3 FA intake prior to diagnosis will have lower grade prostate tumors as determined by Gleason score at diagnosis compared to those men who consume relative lower quantities of ω -3 FA.

Methods: We recruited 172 newly diagnosed men with PCa at the Moffitt Cancer Center, who completed a validated epidemiological, food frequency questionnaire specifically to measure ω -3 fatty acid intake and consented to provide their medical information.

Results: Our results indicated that ω -3 FA intake had no impact on grade at diagnosis of PCa. In the multivariate model, ω -3 FA intake indicated a trend toward higher intake being associated with low Gleason grade after adjusting for age and PSA ($P < 0.25$). A novel observation in this study is that, overall, ω -3 fatty acid intake of all men diagnosed with PCa (mean: 2.8 g /week) in this cohort was significantly lower (75 % lower) than the recommendations of the USRDA for optimal ω -3 fatty acid (11.2 g per week).

Conclusion: With our understanding of the benefits of ω -3 fatty acid intake for overall health, including its role in preventing prostate carcinogenesis, the overall significantly low dietary intake of ω -3 fatty acid in this cohort may be concerning, requiring further education. Additionally, the role of dietary ω -3 fatty acid intake in the modulation of biomarkers of PCa in general, warrants further studies.

1. Introduction

Other than skin cancer, prostate cancer (PCa) is the most common cancer in American men [1]. In 2025, it is estimated that 313,780 new cases of prostate cancer will be diagnosed in the US and 35,770 men will die of this disease ACS [2]. Despite several advances in the early detection, treatment, and clinical management of prostate cancer in recent decades, much work is still needed to fully understand the risk factors for prostate cancer that can result in the development of

strategies to address these risk factors by implementing validated preventive approaches targeting men at high risk or with early-stage disease. Metastatic castration resistant PCa is still a fatal disease [1,2]. Known risk factors for prostate cancer development include age, race, ethnicity, family history, hormonal factors, diet, obesity, and physical activity [3–12].

Several nutrients and nutrient-derived substances have been evaluated as dietary interventions or as bioactive compounds in vitro, pre-clinical and in early clinical trials [13,14]. Some of the compounds and

Abbreviations: FA, Fatty acids; PCa, Prostate Cancer; ω -3 FA, Omega-3 fatty acids; PSA, Prostate specific antigen; EPA, Eicosapentaenoic acid; DHA, Docosahexaenoic acid; NF- κ B, Nuclear factor- κ B; AMPK, Activated protein kinase; SIRT1, Sirtuin 1; Cox, Cyclooxygenase; PTEN, Phosphatase and tensin homolog; ALA, Alpha linolenic acid; PUFA, Polyunsaturated fatty acids; SELECT, Selenium and Vitamin E Cancer Prevention Trial; FFQ, Food frequency questionnaire; USDA, United States Department of Agriculture; BMI, Body mass index; SD, Standard deviation; COVID, Coronavirus Disease; RBC, Red blood cells.

* Correspondence to: MRC-CAN CONT, Moffitt Cancer Center, 13131 Magnolia Drive, Tampa, FL 33612, United States.

E-mail address: Nagi.Kumar@Moffitt.org (N.B. Kumar).

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nutritional approaches evaluated to date include specific diets rich in vegetables and fruits [15], low fat diets [16], vitamins D [17–19], Vitamin E [20,21], green tea catechins [22,23], soy isoflavones [24–26], lycopene [27,28], Selenium [20,29], pomegranate [30], and resveratrol [31,32]. However, to date, the results of these trials have been mixed, potentially due to limitations in validated endpoints, measurement of exposure, target populations, duration of intervention and studies that are underpowered. Thus, currently, there are no nutritional approaches or nutrient-derived agents/bioactive compounds that have been identified as safe and effective to reduce risk or prevent progression of prostate cancer in clinical settings, underscoring the need to continue to identify agents/approaches to reduce risk and prevent progression of PCa.

Omega-3 (ω -3) polyunsaturated fatty acids (PUFA's) -specifically from nutritional sources – eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA), have been shown to exert a variety of anti-tumor effects including inhibition of several proliferative pathways, induction of apoptosis, reduction of angiogenesis and anti-inflammatory effects [33,34]. Research has demonstrated that the mechanisms involved include suppression of nuclear factor- κ B (NF- κ B), activation of AMPK/SIRT1, modulation of cyclooxygenase (COX) activity, and up-regulation of novel anti-inflammatory lipid mediators such as protectins, maresins, and resolvins [35,36]. In a study performed on human prostate adenocarcinoma cell lines found that exposure to DHA attenuated the NF- κ B survival pathway, as well as lowered levels of survivin, a protein involved in inhibition of apoptosis, thus arresting cancer growth [37]. ω -3 FA and ω -6 FA have opposing roles in contributing to inflammatory pathways. The main dietary ω -3 FA include alpha linolenic acid (ALA), DHA, and EPA. These molecules or their derivatives are modulated by cyclooxygenase (COX) enzymes, which also are responsible for enzymatic reactions on arachidonic acid, derived from omega-6 FA. Unlike ω -3 FA, the omega-6 pathway produces various pro-inflammatory molecules. A study targeting PTEN knockout mice found that diets high in ω -3 FA in the mouse model were able to promote apoptosis by increasing cellular levels of BAD (a pro-apoptotic protein), lowering its phosphorylation, and significantly increased survival of the PTEN knockout mice over a 12-month period [38]. Other have shown that ω -3 FA enriched diet decreased prostate TRAMP-C2 tumor growth in immune-competent eugonadal and castrated mice demonstrating that ω 3 FA can inhibit tumor cell growth and induce a local anti-tumor inflammatory response, independently of androgen levels [39].

Epidemiologic studies have shown that populations that consume higher amounts of marine-derived ω -3 FA, including Japanese and Alaskan Natives, have lowered risk for the development of PCa [40]. In a prospective cohort of 47 866 US men aged 40–75 followed for 14 years, results demonstrated a decreased risk of PCa with ω -3 FA intake [41]. In a recent meta-analysis, the dose-response association between fish intake and prostate cancer was determined. In the dose-response analyses, each 20 g/day increase in total fish intake was associated with a 12 % lower risk of prostate cancer mortality, supporting the protective association between total fish intake and the risk of prostate cancer mortality [42]. On the other hand, in a prospective trial [SELECT] [43], high plasma phospholipid concentrations of long-chain ω -3 PUFA were associated with statistically significant increases both low- and high-grade disease and for EPA, DPA and DHA, although, these ω -3 FA have been shown to be anti-inflammatory, metabolically interrelated ω -3 fatty acids derived from oily fish and fish oil supplements. Comparable results have been reported in several meta-analysis with three supporting the observations in the SELECT cohort with another reporting an inverse association [44–46]. Among interventional trials in PCa, prostate specific antigen (PSA) levels remained unchanged despite fish oil supplementation; however, in 2 of these interventional trials, reductions in either cell-cycle progression score or reduction in epithelial cell proliferation were noted [34]. In a randomized clinical trial from biopsy to prostatectomy (n = 48), a 4–6-week low-fat diet and 5 g fish oil capsules/day to achieve an ω -6: ω -3 FA ratio of 2:1 resulted in decreased

prostate cancer proliferation and decreased prostate tissue omega-6: omega-3 ratios [47]. In a study to evaluate associations between grade reclassification and ω 3 levels assessed in prostatic tissue, red blood cells (RBC), and diet. Results indicated that high long-chain (LC) ω 3-eicosapentaenoic acid (EPA) level in prostate tissue and LC ω 3 dietary intake was associated with lower odds of high-grade PCa. LC ω 3-EPA levels in the target prostate tissue as well as intake are inversely associated with high-grade PCa in men with low-risk PCa [48]. In a meta-analysis evaluating the association of n-3 FA and PCa risk, the data was reported to be insufficient to suggest a relationship between fish-derived ω -3 fatty acid and risk of PCa [49]. Others have reported comparable results [50].

Despite promising laboratory and pre-clinical data, the overall evidence for the ability of ω -3 fatty acid intake to prevent PCa development, severity, and progression remains inconclusive due to the conflicting results of studies in humans. Based on laboratory and pre-clinical data and lack of clear evidence in observational studies, we hypothesize that patients with higher intake of ω -3 fatty acid prior to diagnosis will have lower grade prostate tumors as determined by Gleason score at diagnosis compared to those men who consume relative lower quantities of ω -3 FA. The hypothesis was tested using the following specific aims. In newly diagnosed men with prostate cancer (between 2020 and 2021), we obtained dietary and supplementary use of ω -3 fatty acid in the past 6 months, using a validated food frequency questionnaire, and evaluated if quantity of intake of ω -3 FA is associated with grade of prostate cancer at diagnosis, taking into account other comorbidities known to be associated with prostate cancer risk (race, age, obesity and other life-style factors).

2. Materials and methods

2.1. Research methods

The study was approved by the Scientific Review Committee at the Moffitt Cancer Center, Tampa, Florida and the Institutional Review Board, Advarra (6100 Merriweather Drive, Columbia, MD 21044) on 18th May, 2021 and received exempt status (IRB# 00000971) determination. In this retrospective cohort study, we identified and recruited consecutive prostate cancer patients admitted to the Moffitt Cancer center Genito-urinary Oncology program, between January 2020 - December 2021 who were newly diagnosed PCa patients. Men who had no previous history of all other cancers, including PCa were included in this study. We contacted these subjects first via mail by forwarding an information letter stating the purpose of the study and obtaining consent via mail to participate in addition to a questionnaire package containing (a) a demographic questionnaire and a (b) validated food frequency questionnaire specifically for obtaining ω -3 PUFA. Subjects were requested to return the questionnaires within the next 10 days of receipt in a self-addressed stamped envelope that was provided with the package. We followed the guidelines as outlined by Readex (Don Dillman's Total Survey Design Method) (<https://www.readexresearch.com>) to maximize response rates After the mailing of the initial letter and questionnaire package to the potential subjects, we followed up with an alert letter to ensure that the package was received. Non-responders to the follow up alert letter were contacted via telephone call after 2 weeks. All subjects who completed the questionnaires were PCa patients, first diagnosed within the past 18 months. We obtained permission from the subjects to abstract data from their medical records for this study, which included the following data from subject's electronic medical records: Gleason score (primary endpoint) and serum PSA. For purposes of this study, we considered Gleason 3 + 3 and 3 + 4 as low-grade tumors and Gleason 4 + 3 and above as high-grade tumors. We obtained epidemiological data using a Demographic Questionnaire, which included age, race, ethnicity, anthropometrics (Height, Weight) family history of cancer including prostate cancer, history of environmental and personal exposures, alcohol intake and physical activity, cancer screening,

medication use including nutritional supplements and co morbidities. We have used this questionnaire in our previous trials targeting this patient population [22,25,31,32]. In addition, we included a validated Food Frequency Questionnaire (FFQ) to assess ω -3 foods and supplements intake, previously validated by Shen et al. [51]. The FFQ is a directed dietary questionnaire that aims to quantify intake levels of both supplement-based and marine-based dietary ω -3 PUFA's. The FFQ was validated by the U.S. EPA Human Studies Facility via a study aiming to estimate the protective effects of ω -3 PUFA's against air pollution related cardiovascular and pulmonary dysfunctions [51]. This questionnaire queries the subject about seafood type, serving, frequency, intake of ω -3 fatty acid fortified foods, and ω -3 fatty acid supplement use including frequency, brand, and dose over the prior 6 months before presentation. The intake of ω -3 FA via supplement use was calculated based of brand label information, while the dietary intake from various food sources was calculated based on a USDA database [52] detailing the amounts of ω -3 FAs in each food group outlined by the questionnaire. In addition to quantifying exposure using this FFQ, these results were validated utilizing a blood-based biomarkers, ω -3 index [53] and whole blood levels of EPA and DHA. ω -3 index is a long-term, stable marker of ω -3 FA status that assesses the amount of EPA (eicosapentaenoic acid) and DHA (docosahexaenoic acid) in red blood cell (RBC) [53]. Statistical analyses were performed to measure the level of concordance between self-reported ω -3 FA intake from the FFQ described with the ω -3 FA index as well as whole blood levels of EPA and DHA data. The authors found the weekly consumption of EPA and DHA as calculated from the FFQ was well correlated with EPA, DHA, and ω -3 FA ($r = .67, .62, .67$, $p < .001$). The relationship between high intake of EPA and DHA with omega-3 index was as follows: sensitivity 89 %, specificity 84 %, and agreement 86 %. The relationship between low ω -3 FA intake of EPA and DHA with ω -3 FA index was as follows: sensitivity 100 %, specificity 66 %, and agreement 42 %. Thus, the authors concluded that the questionnaire is a valid screening tool for individuals with high and low ω -3 FA index, especially those with high index. Additionally, using a tailored food-frequency questionnaire to assess ω -3 FA has been observed to be better correlated to determine ω -3 PUFA intakes than the relatively long and more comprehensive FFQ [54–56]. Data obtained using a 7-question screener designed to capture DHA intake (DHA Food Frequency Questionnaire, DHA-FFQ) was highly correlated with RBC-DHA ($rs=0.528$) than a full dietary questionnaire (DHQ-II) ($rs=0.352$) [56]. Considering that our study aims to observe the association primarily between high ω -3 FA intake and risk for grade PCa as defined by Gleason score, this questionnaire is an optimal instrument for determining ω -3 FA intake.

Using the validated, tailored FFQ [51,54–56], we quantified ω -3 FA intake over the prior 6 months before diagnosis, specifically as a weekly gram intake of DHA and EPA and compared these data to the pathology-confirmed Gleason score. The primary outcome was the effect of ω -3 FA intake on grade of PCa, as indicated by Gleason score. Based on the frequency of the data, categorization of ω -3 FA intake was as follows: None (0.0 g/week EPA+DHA), low ≤ 3 g/week, and high (>3 g/week). All data were entered into a database specifically designed for this study. All data were entered into a database specifically designed for this study. Subjects were de-identified once all data was abstracted and data analysis was complete. All data will be retained at the Moffitt Cancer Center. Subjects were not identified in any of the publications or presentations evolving from this study. No formal power calculation was done at the planning stage due to the retrospective design and limited prior data.

2.2. Statistical methods

The primary analysis was logistic regression, with presence of high-grade prostate cancer (Gleason score of 4 + 3, 4 + 4 or higher) versus low-grade prostate cancer (Gleason score of 3 + 3 and 3 + 4) being the primary outcome. The independent variable of most interest is the

intake of weekly ω -3 FA ((EPA+DHA)/week) from dietary sources and supplements, measured continuously using a square-root scale. Additional independent variables included were age at diagnosis, race, family history of cancer, smoking and alcohol use. We also built a second logistic regression model, adding clinical variables including body mass index (BMI) and serum prostate specific antigen (PSA). The rationale for including two models is that the ω -3 FA consumption can influence both BMI and PSA, thereby shifting the odds ratios associated with the ω -3 FA dummy variables. Thus, our plan had 6 independent variables in model 1 and 8 in model 2. We anticipated that we would have ~500 responses to the questionnaire, so that we would have ~70 patients in the smaller Gleason score group, to enable us to examine up to 14 independent variables (to provide a minimum of 5 events per variable in the smaller response group). While we only had 172 respondents, there were 58 in the smaller response group, satisfying the minimum of 5 events per variable. Demographic and clinical variables were summarized as mean and standard deviation (SD) for continuous variables and as frequencies and proportions for categorical variables. The association between grade of prostate cancer as indicated by Gleason score and other variables was evaluated using Fisher's exact test or the Chi-square test for categorical variables, and the Wilcoxon rank-sum test for numerical variables. A multivariable logistic regression model was used to assess the association between prostate cancer grade and explanatory variables. Model development began by including all demographic and clinical variables in the initial model, followed by backward elimination to remove variables with $p > 0.25$, resulting in the final model. The choice of a p-value-to-stay in the backward elimination multivariable model is not standardized; our choice of $p = 0.25$ was recommended by Mickey and Greenland [57]. All analyses were performed using SAS (Version 9.4, SAS Institute Inc., Cary, NC). The reporting of this study conforms to STROBE guidelines [58].

3. Results

Of the total newly diagnosed PCa patients, 664 were eligible for the study of which 172 responded with completed surveys (26 %). Of the eligible subjects, 114 (66 %) men were diagnosed with low-grade PCa at diagnosis and 58 (34 %) were diagnosed with high-grade tumors. The survey response rate of 26 % was lower than anticipated but linked to the new environmental challenges caused by the COVID-19 epidemic and the additional of natural disasters that impacted Florida and the infrastructure our patient population [59]. As noted, no formal power calculation was done at the planning stage due to the retrospective design and limited prior data. A post-hoc power analysis using G*Power (version 3.1.9.7) showed that with 172 respondents (58 in the smaller Gleason score group, 114 in the larger), the study had ~82 % power to detect a medium effect size (Cohen's $d = 0.5$) at $\alpha = 0.05$, two-tailed. Looking for possible response bias, we observed that 115/116 patients with PSA < 1 were non-respondents. For patients with PSA levels ≥ 1 , the response rate was 31 % and no statistically significant difference were noted by age at diagnosis, Gleason score, or PSA level (Supplementary Table 1). Table 1 displays the baseline characteristics of all study participants. The 2 study arms were well matched for potential predictive markers, including ethnicity, race, marital status, education, height, weight, body mass index, tobacco use and alcohol use. Age at time of diagnosis was significantly higher for higher grade tumors (P value < 0.001). Table 2 displays clinical variables of study subjects by grade of PCa. Serum prostate specific antigen (PSA) was higher in men with higher grade tumors as anticipated ($P = 0.001$). Men with high-grade tumors had a statistically higher concentration of serum PSA (> 10 ngs/mL) ($P < 0.05$) compared to men with low-grade tumors. A greater number of men in both groups had a family history of PCa. Finasteride, Aspirin or Statin use, and other comorbidities did not impact grade of cancer at diagnosis. Table 3 displays ω -3 FA intake in study subjects and grade of PCa. Low or relatively higher ω -3 FA intake had no impact on grade at diagnosis of PCa. However, in the multivariate model

Table 1

Lifestyle characteristics of the study subjects by grade of prostate cancer (N = 172, row percent).

Variables	Total N = 172 N (%)	Low Grade Gleason 3 + 3 and 3 + 4 N = 114 (66 %) N (%)	High Grade Gleason $\geq$ 4 + 3 N = 58 (34 %) N (%)	p-value*
Age at recruitment (years)				< 0.001
Mean (SD)	65.7 (8.3)	64.1 (8.5)	68.8 (6.9)	
Ethnicity				0.968
White	150	98 (65 %)	52 (35 %)	
Black/African American	13	9 (69 %)	4 (31 %)	
Asian	4	3 (75 %)	1 (25 %)	
Other	5	4 (80 %)	1 (20 %)	
Race				1.000
Non-Hispanic	157	104 (66 %)	53 (34 %)	
Hispanic	7	5 (71 %)	2 (29 %)	
Unknown	8	5 (62 %)	3 (38 %)	
Marital Status				0.342
Single	9	4 (44 %)	5 (56 %)	
Married	147	100 (68 %)	47 (32 %)	
Cohabiting/living together	2	2 (100 %)	0 (0 %)	
Divorced/separated	9	6 (67 %)	3 (33 %)	
Widowed	5	2 (40 %)	3 (60 %)	
Education				0.739
High school graduate/GED or below	18	11 (61.1 %)	7 (38.9 %)	
Vocational school/Some college	36	26 (72.2 %)	10 (27.8 %)	
Graduated college	51	32 (62.7 %)	19 (37.3 %)	
Postgraduate or professional school	65	45 (69.2 %)	20 (30.8 %)	
Unknown	2	0 (0 %)	2 (100 %)	
Height at diagnosis (inches)				0.864
Mean (SD)	69.7 (2.5)	69.7 (2.5)	69.8 (2.4)	
Weight at diagnosis (lbs)				0.706
Mean (SD)	199 (31.5)	199.5 (31.8)	198.3 (31.3)	
BMI (kg/m²)				0.734
Mean (SD)	28.8 (4.5)	28.9 (4.5)	28.7 (4.5)	
Smoking status, N (%)				0.376
Current	3	1 (33 %)	2 (67 %)	
Former	67	43 (64 %)	24 (36 %)	
Never	102	70 (69 %)	32 (31 %)	
Pack- years (Packs/day x No. years)				0.629
Mean (SD)	12.8 (13.2)	12.1 (12.6)	14.1 (14.4)	
Alcohol				0.678
No	30	21 (70 %)	9 (30 %)	
Yes	142	93 (66 %)	49 (34 %)	

(Table 4), ω -3 FA intake indicates a trend toward higher intake being associated with low Gleason grade after adjusting for age and PSA ($P < 0.25$). Fig. 1 displays the distribution of the square root of weekly ω -3 FA intake by Gleason score and indicates a trend to-wards greater number of subjects diagnosed with low-grade tumors had a relatively higher intake of ω -3 FA; however, these results were not statistically significant. Similarly, Fig. 2 displays the distribution of the square root of weekly ω -3 FA intake by serum PSA and indicates a trend to-wards greater number of subjects diagnosed with lower serum PSA had a relatively higher intake of ω -3 FA; however, these results were not statistically significant.

4. Discussion

In this study evaluating the impact of weekly intake of ω -3 fatty acid intake 6 months prior to first diagnosis of PCa, we did not observe a statistically significant association between intake of ω -3 fatty acid and grade of PCa. Using a multivariate model, a higher intake of ω -3 FA intake showed a trend for an association with lower grade tumors as indicated by Gleason score, after adjusting for age and serum PSA ($P < 0.25$). The US recommended dietary allowance for n-3 FA for adult men is 1.6 g a day or 11.2 g per week. However, the mean intake of this patient cohort with low-grade PCa was 3.1 g (SD 5.6) per week and for high-grade PCa, 2.2 g (SD 3.1) per week, substantially lower than the recommendations of the USRDA for optimum ω -3 fatty acid intake. Thus, our finding may be largely driven by the overall suboptimal intake

of ω -3 FA by this cohort of men newly diagnosed with PCa. Consequently, these findings should be interpreted with caution. Additionally, we observed that age at time of diagnosis was significantly higher for higher grade tumors (P value < 0.001), and serum PSA was higher in higher grade tumors ($P = 0.001$).

Previous studies have used various strategies to collect exposure data to ω -3 FA. Plasma, RBC and adipose tissue have been utilized to quantify exposure to ω -3 fatty acids over time. With limited access to blood samples or prostate tissue in specific population studies, exposure data to ω -3 fatty acids have been obtained using FFQ, although subject to recall bias. Similar to the work completed by Shen et al. [51] who validated the data obtained from FFQ with blood-based biomarkers (ω -3 FA index and plasma levels of ω -3 fatty acids), more recently, several groups have validated data obtained from FFQ with biomarkers of exposure. Studies have shown that data collected using a web-FFQ is an accurate tool to assess total long-chain ω -3 FAs, EPA, and DHA but not ALA intake in clinical trials and epidemiological studies in men with PCa [60]. In a study that analyzed ω -3 fatty acids from the red blood cell (RBC) membranes and ω -3 fatty acids intakes using FFQ at baseline, researchers observed significant correlations between fatty fish intake and total ω -3 FAs in RBC membranes [60]. Similar results have been observed in other studies validating intake of ω -3 FAs using FFQ compared with ω -3 FAs in RBCs [51,54–56]. In a large, prospective trial [SELECT] [43], high plasma phospholipid concentrations of long-chain ω -3 PUFA were obtained as a biomarker of ω -3 fatty acids exposure derived from oily fish and fish oil supplements. However, this was a

Table 2

Clinical variables of the study subjects by grade of prostate cancer.

Variable	Total N = 172	Low -Grade Gleason 3 + 3 and 3 + 4 N = 114 (66 %)	High- Grade Gleason $\geq 4 + 3$ N = 58 (34 %)	p- value
PSA (ng/mL)	N %	N %	N %	
Mean (SD)	8.0 (8.2)	6.6 (3.4)	10.7 (13.0)	0.001
Median (Q1–Q3)	6.1 (4.7 – 8.1)	5.5 (4.5 – 7.7)	7.0 (5.4 – 11.0)	
PSA (ng/mL)				
≤ 10	138	96 (70)	42 (30)	0.040
> 10	30	15 (50)	15 (50)	
Unknown	4	3 (75)	1 (25)	
Gleason Score:				
6(3 +3)	61	61 (100 %)	0 (0 %)	0.223
7(3 +4)	53	53 (100 %)	0 (0 %)	
7(4 +3)	24	0 (0 %)	24 (100 %)	
$\geq 8(4 +4)$	34	0 (0 %)	34 (100 %)	
Other diseases				
Diabetes (Y)	12	6 (50 %)	6 (50 %)	0.590
Prostate problems (Y)	124	84 (68 %)	40 (32 %)	
Medications Supplements				
Finasteride use (Y)	3	1 (33 %)	2 (67 %)	0.753
Aspirin use (Y)	11	8 (73 %)	3 (27 %)	
Statin (Y)	1	0 (0 %)	1 (100 %)	
Family history of cancer				0.369
Yes	126	86 (68 %)	40 (32 %)	
No	46	28 (61 %)	18 (39 %)	

one-time plasma ω -3 fatty acids measurement to account for exposure during the data collection period. There was no other data available on tissue markers of ω -3 FA nor actual nutritional intake of ω -3 FA. In a recent study that explored the relationship between dietary intake of ω -3 FA and prostate cancer grade, researchers measured exposure using RBC concentrations of ω -3 FA, prostate tissue levels as well as nutritional intake of ω -3 FA using a FFQ. Dietary intake of ω -3 FAs in this study was consistent with prostate tissue levels in terms of its implication for disease progression, which supports the validity of analyzing the dietary intake of ω -3 FAs to determine risk for PCa. RBC concentrations of ω -3 FA was not correlated to nutritional intake, prostate tissue levels nor associated to grade of PCa [48]. Similar observations have been made in intervention trials where intake has been correlated markers in tissue biomarkers - cell-cycle progression score or reduction in epithelial cell or prostate cancer proliferation [47]. It is clear that to reduce recall bias and the complexities of using plasma or tissue markers of exposure, future prospective studies should continue to develop and validate composite biomarkers to quantify nutritional intake of ω -3 FA [48].

As with all retrospective, cohort studies, there were a few limitations

to this study. First, we had no prespecified power calculation. We followed the guidelines as outlined by Readex (Don Dillman's Total Survey Design Method) (<https://www.readexresearch.com>) to maximize response rates. We forwarded an initial letter and questionnaire package to the potential subjects and followed up with an alert letter with a follow-up survey and phone call to non-responders after 2 weeks. All questionnaires included address and postage paid envelopes. However, we did not provide a monetary incentive due to lack of funding. The relatively lower response rate was linked to the new environmental challenges caused by the COVID-19 epidemic and the additional of natural disasters (2 major category hurricanes) that impacted Florida

Table 4

Multivariable Logistic Regression Analysis of Significant Variables that were predictive of risk for high-grade prostate cancer.

Covariate	Level	Gleason Score= 'High Grade'	
		Odds Ratio (95 % CI)	P-value
Age at diagnosis		1.08 (1.03–1.13)	< .001
ω -3 fatty acid intake (Square Root)		0.83 (0.61–1.13)	0.231
PSA at diagnosis	> 10	2.29 (0.98–5.37)	0.057
	≤ 10	-	-

* Number of observations in the original data set = 172. Number of observations used = 163.

* All predictive variables were added to the initial model, and then backward selection was applied. Variables with $p > 0.25$ were dropped from the final model.

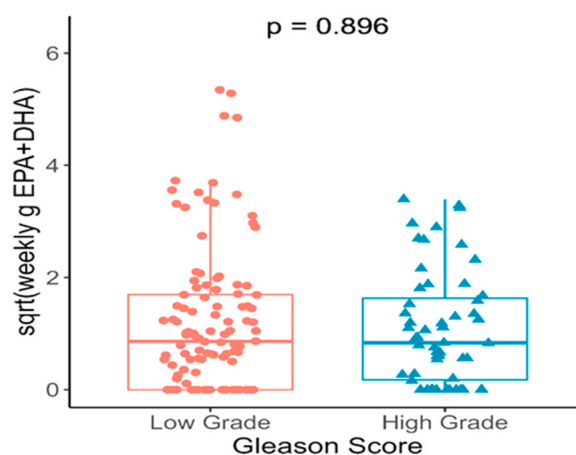


Fig. 1. Distribution of the Square Root of Weekly Omega-3 Fatty Acid Intake (grams) by Gleason Score at Diagnosis. This boxplot illustrates the distribution of omega-3 fatty acid intake (EPA + DHA) by Gleason Score at diagnosis, categorized into Low-Grade and High-Grade groups. The boxplot displays the median and interquartile range, while jittered points represent individual data points. The median omega-3 intake appears similar between the Low-Grade and High-Grade groups.

Table 3Grade of prostate cancer by ω -3 fatty acid intake (grams)/week of study subjects.

Fatty Acid	Total N = 169	Low- Grade Gleason 3 + 3 and 3 + 4 N = 112	High Grade Gleason $> 4 + 3$ N = 57	OR (95 % CI)	p-value
	N %	N %	N %		
ω-3 fatty acid intake					
Mean (SD)	2.8 (5.0)	3.1 (5.6)	2.2 (3.1)		0.896
None	42	29 (69)	13 (31)	Reference	
Low/Intermediate ≤ 3 g	86	55 (64)	31 (36)	1.26 (0.57–2.77)	0.808
High: > 3 g	41	28 (68)	13 (32)	1.04 (0.41–2.62)	

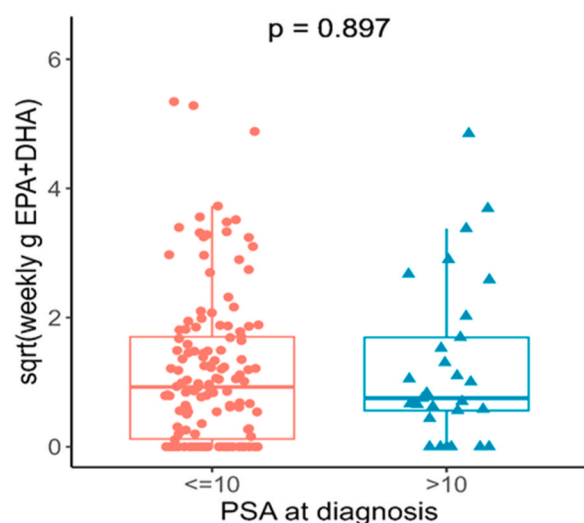


Fig. 2. Distribution Square root of Weekly ω -3 fatty acid intake (grams) by Serum PSA (ng/mL) at Diagnosis. Distribution of omega-3 FA intake (EPA + DHA) by PSA levels at diagnosis (≤ 10 vs >10). The boxplot shows the median and interquartile range, while jittered points represent individual data points. The median omega-3 intake is slightly higher for PSA ≤ 10 compared to PSA >10 .

and the infrastructure our patient population [59]. Several of our patients were in addition displaced from their homes during the hurricanes for several months and much of the mail service was disrupted in the State. However, when patients with PSA levels ≥ 1 , were examined for age, Gleason score, or PSA level, no statistically significant differences were seen. Future studies should use larger samples and prespecified power calculations.

Additional limitations of our study are the lack of biomarkers such as ω -3 FA index or prostate tissue concentrations of ω -3 FAs that has previously been shown to be a valid biomarker of exposure to ω -3 FAs [48], in addition to self-reported nutritional and supplement intake of ω -3 FAs. The use of a single dietary screening tool may introduce measurement error compared with biomarker assays which have been shown to be more stable over time [53]. However, due to the lack of availability of tissue from PCa biopsy or plasma samples and costs of testing, we were unable to determine tissue or plasma concentrations of ω -3 FA. The food frequency questionnaire was also limited to obtaining only ω -3 FA intake. Since the biology of ω -3 FA fatty acids intake is intrinsically related to that of ω -6 FA, it may be critical to quantify both these fatty acid intakes. There was a lack of diversity in the responders, where the predominant population were White men. These are limitations that should be considered to improve the rigor of research designs in future studies.

5. Conclusions

In summary, a novel observation in this study is that, overall, ω -3 fatty acid intake of all men diagnosed with PCa (mean: 2.8 g /week) in this cohort was significantly lower (75 % lower) than the recommendations of the USRDA for optimal ω -3 fatty acid (11.2 g per week). Early studies examining the role of ω -3 FA in prostate cancer development and progression are mixed and inconclusive potentially due to significantly low intake of ω -3 FA in this population, lack of comprehensive and systematic evaluation of intake from nutritional and supplement sources, utilizing a validated questionnaire to specifically measure ω -3 FA intake and taking into consideration potential confounders. Additionally, the validity of the use of RBCs to determine ω -3 FA intake has been in question, as this marker may reflect metabolic activity of a plethora of tissues. However, with our understanding of the mechanism of action of ω -3 FA in prostate carcinogenesis from laboratory studies and preclinical

data and some promising early indications from human studies, the correlation between ω -3 FA intake and risk for development of PCa in general, warrants further studies. Future studies should include prospective research designs that include serial measurements of nutritional and supplement intake ω -3 FA intake, tissue concentrations of ω -3 FA along with other novel tissue and plasma biomarkers that can improve the rigor of the current research methodologies and strengthen these observations.

CRedit authorship contribution statement

Junmin Whiting: Data curation, Formal analysis, Writing – review & editing. **Schell Michael:** Conceptualization, Data curation, Formal analysis, Investigation, Writing – review & editing. **Michael Poch:** Data curation, Formal analysis, Resources, Writing – review & editing. **Brandon Manley:** Investigation, Writing – review & editing. **Julio Pow-Sang:** Investigation, Writing – review & editing. **Alice Yu:** Investigation, Writing – review & editing. **Kumar Nagi:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing. **Daniel Lemay:** Data curation, Formal analysis, Writing – review & editing. **Saira Bahl:** Data curation, Formal analysis, Writing – review & editing. **Jasreman Dhillon:** Data curation, Formal analysis, Investigation, Writing – review & editing. **Roger Li:** he participated in the review of the data and data interpretation as well as review and revision of the manuscript.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Nagi B Kumar reports financial support was provided by Pinellas Partners. Nagi B Kumar reports a relationship with Moffitt Cancer Center that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.canep.2025.102959](https://doi.org/10.1016/j.canep.2025.102959).

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